

Test Execution Report

Project: QA Test Web App – Automation Suite **Application under test:** <https://qa-test-web-app.vercel.app/index.html> **Framework:** Playwright v1.58.2 | JavaScript (ESM) **Pattern:** Page Object Model (BasePage.js) **Executed by:** Karlo Marić **Execution date:** 2026-02-22 **Environment:** Windows 11 Pro | Node.js | Chromium (Desktop Chrome)

1. Executive Summary

A total of **14 automated end-to-end tests** were executed against the QA Test Web App. The suite covers registration, authentication, form validation, navigation, and session management.

11 tests passed and **3 tests failed intentionally** — each failing test was written specifically to expose and document a known defect in the application. No production-level regressions were found. All functional flows work as expected.

2. Test Metrics

Metric	Value
Total tests	14
Passed	11
Failed	3
Skipped	0
Flaky	0
Pass rate	78.6%
Fail rate	21.4%
Total execution time	~1 min 30 sec
Execution date	2026-02-22
Browser	Chromium
Workers	1 (sequential)
Retries on failure	0

Test type distribution

Type	Count	% of total
Functional	6	42.9%
Negative	4	28.6%
E2E	1	7.1%
Bug	3	21.4%

Result by test type

Type	Passed	Failed
Functional	6	0
Negative	4	0
E2E	1	0
Bug	1*	3

*TC-007 (MismatchedPasswordsTest) is counted as passed because it asserts that the buggy behavior actively occurs — confirming the bug. The 3 failed bug tests are intentionally designed to fail to signal the defect.

3. Test Execution Results

#	Test ID	Test Name	Type	Result	Order
1	TC-011	Registration form fields cleared after page refresh	Functional	 PASS	1/14
2	TC-006	Cannot create account with existing email	Negative	 PASS	2/14
3	TC-005	Register and login with new account	E2E	 PASS	3/14
4	TC-003	User can log out after logging in	Functional	 PASS	4/14
5	TC-004	User is redirected to the Dashboard after login	Functional	 PASS	5/14

#	Test ID	Test Name	Type	Result	Order
6	TC-002	Login with existing account	Functional	✓ PASS	6/14
7	TC-010	Login with invalid password shows error	Negative	✓ PASS	7/14
8	TC-007	[BUG] Mismatched passwords accepted at registration	Bug	✓ PASS	8/14
9	TC-012	[BUG] Redirected to Login page after registration	Bug	✗ FAIL	9/14
10	TC-009	Register with invalid email (no @ symbol)	Negative	✓ PASS	10/14
11	TC-001	Register a new account	Functional	✓ PASS	11/14
12	TC-013	[BUG] Required fields not marked with asterisk	Bug	✗ FAIL	12/14
13	TC-008	Submit empty fields shows validation warnings	Negative	✓ PASS	13/14
14	TC-014	[BUG] App accepts 4-character weak password	Bug	✗ FAIL	14/14

4. Detailed Results for Failing Tests

TC-012 – [BUG] User is redirected to Login page after registration

File: tests/RedirectingAfterRegistrationTest.spec.js **Status:** FAIL **Failure type:** Intentional – documents a confirmed bug

Expected behavior: After a successful registration, the user should be automatically logged in and redirected to the **Dashboard** (/dashboard.html).

Actual behavior: The app redirects the user back to the **Login page** (/index.html?registered=true).

Failing assertion:

```
Error: expect(page).not.toHaveURL(expected) failed
```

```
Expected pattern: not /index\.html/
```

```
Received string: "https://qa-test-web-app.vercel.app/index.html?registered=true"
```

Location: tests/RedirectingAfterRegistrationTest.spec.js:42 **Screenshot evidence:** bug-redirected-to-login.png

TC-013 – [BUG] Required fields are not marked with an asterisk

File: tests/RequiredFieldsAsteriskTest.spec.js **Status:** FAIL **Failure type:** Intentional – documents a confirmed bug

Expected behavior: All required fields on the registration form (First Name, Last Name, Email Address, Phone Number, Street Address, City, ZIP Code, Password, Confirm Password) should display an asterisk (*) next to their label to indicate they are mandatory.

Actual behavior: None of the field labels contain an asterisk. The first checked label "First Name" is returned as-is with no * character.

Failing assertion:

```
Error: Expected "First Name" label to contain an asterisk (*)

Expected substring: "*"
Received string:    "First Name"
```

Location: tests/RequiredFieldsAsteriskTest.spec.js:32 **Screenshot evidence:** required-fields-asterisk.png

TC-014 – [BUG] App accepts a password with only 4 characters

File: tests/WeakPasswordTest.spec.js **Status:** FAIL **Failure type:** Intentional – documents a confirmed bug

Expected behavior: The application should reject passwords shorter than 8 characters and display a validation message such as "Password must be at least 8 characters".

Actual behavior: The app accepts a 4-character password (abcd) and proceeds to register the account successfully, redirecting the user to the login page with ?registered=true .

Failing assertion:

```
Error: expect(locator).toBeVisible() failed

Locator: locator('text=/password must be at least 8/i')
         .or(locator('text=/password too short/i'))
         .or(locator('text=/minimum.*8.*characters/i'))
Expected: visible
Timeout: 5000ms
Error: element(s) not found
```

Location: tests/WeakPasswordTest.spec.js:44 **Screenshot evidence:** weak-password-form.png , weak-password-result.png

5. Bug Summary

Bug ID	Severity	Title	Test	Status
BUG-001	Medium	Wrong redirect after registration (Login instead of Dashboard)	TC-012	Open
BUG-002	High	Mismatched passwords are accepted during registration	TC-007	Open
BUG-003	Low	Required fields have no asterisk (*) visual indicator	TC-013	Open
BUG-004	High	Weak 4-character password accepted without validation	TC-014	Open

Severity explanation

- **BUG-001 (Medium):** Post-registration UX is broken — user must manually log in after registering. Functional but causes friction.
- **BUG-002 (High):** Security concern — users can register with a mismatched Confirm Password. The app silently uses the Password field value and creates the account. No error is shown.
- **BUG-003 (Low):** UX/accessibility issue — users have no visual hint of required fields before attempting to submit.
- **BUG-004 (High):** Security concern — passwords as short as 4 characters are accepted. Minimum password length should be enforced server-side and client-side.

6. Test Coverage

Covered user flows

Flow	Covered
New user registration (happy path)	✓
Registration with invalid email	✓
Registration with duplicate email	✓
Registration with mismatched passwords	✓

Flow	Covered
Registration with weak password	✓
Registration with empty fields	✓
Login (happy path)	✓
Login with wrong password	✓
Login → Dashboard redirect	✓
Register → Login flow (E2E)	✓
Logout	✓
Form data persistence on refresh	✓
Post-registration redirect behavior	✓
Required field visual indicators	✓

Not covered (out of scope / not present in app)

Area	Reason
Password reset / forgot password	Feature not present in app
Email verification	Feature not present in app
Profile editing	Feature not present in app
Cross-browser (Firefox, WebKit)	Run on Chromium only
Mobile viewports	Out of scope for this sprint
API-level tests	Out of scope

7. Sprint Tasks

The following automation tasks were completed as part of this sprint:

#	Task	Status
1	Set up Playwright project with POM structure	✓ Done
2	Implement BasePage with shared helper methods	✓ Done
3	Implement test data management (credentials.js)	✓ Done
4	Write registration happy path test (TC-001)	✓ Done

#	Task	Status
5	Write login happy path test (TC-002)	✓ Done
6	Write login → logout flow test (TC-003)	✓ Done
7	Write Dashboard redirect test (TC-004)	✓ Done
8	Write register → login E2E test (TC-005)	✓ Done
9	Write duplicate email negative test (TC-006)	✓ Done
10	Write mismatched passwords bug test (TC-007)	✓ Done
11	Write empty fields validation test (TC-008)	✓ Done
12	Write invalid email negative test (TC-009)	✓ Done
13	Write wrong password negative test (TC-010)	✓ Done
14	Write page refresh data cleared test (TC-011)	✓ Done
15	Write post-registration redirect bug test (TC-012)	✓ Done
16	Write required fields asterisk bug test (TC-013)	✓ Done
17	Write weak password bug test (TC-014)	✓ Done
18	Capture screenshots at key test steps	✓ Done
19	Run full test suite and validate results	✓ Done
20	Write README with setup instructions	✓ Done
21	Write test execution report with metrics	✓ Done

8. Screenshots Evidence

All screenshots are saved in the project root directory and correspond to key steps in each test.

Screenshot	Test	Description
create-account.png	TC-001	Registration form (empty state)
form-filled.png	TC-001	Registration form with valid data
registration-success.png	TC-001	"Registration successful" message displayed
first-registration.png	TC-006	First registration in duplicate email test

Screenshot	Test	Description
duplicate-email-error.png	TC-006	"Already exists" error on second registration
login-filled.png	TC-002	Login form with credentials entered
after-login.png	TC-002	Dashboard after successful login
login-redirect-before.png	TC-004	Login page before submitting credentials
login-redirect-after.png	TC-004	Dashboard after redirect from login
logout-before.png	TC-003	Dashboard with Logout button visible
logout-after.png	TC-003	Login page after logout
empty-fields-before-submit.png	TC-008	Registration form with only First Name + Email
empty-fields-after-submit.png	TC-008	Browser validation triggered on empty fields
invalid-email-create-account.png	TC-009	Registration form (empty)
invalid-email-form-filled.png	TC-009	Form with invalid email (no @)
invalid-email-submitted.png	TC-009	"Invalid email address" error shown
invalid-password-filled.png	TC-010	Login with valid email, wrong password
invalid-password-result.png	TC-010	"Invalid email or password" error banner
mismatched-passwords-form.png	TC-007	Registration with different password values
mismatched-passwords-after-register.png	TC-007	App redirects to login (no error shown)
mismatched-passwords-login-result.png	TC-007	Successful login despite mismatched passwords
form-before-refresh.png	TC-011	Form partially filled before refresh

Screenshot	Test	Description
form-after-refresh.png	TC-011	Form cleared after page reload
required-fields-asterisk.png	TC-013	Registration form — no asterisks on labels (BUG)
weak-password-form.png	TC-014	Form with 4-char password submitted
weak-password-result.png	TC-014	No validation error shown (BUG)
bug-redirected-to-login.png	TC-012	After registration — redirected to Login page (BUG)

9. Recommendations

1. **Fix BUG-002 (High):** Add server-side and client-side validation to ensure `Password === Confirm Password` before account creation. This is a data integrity issue.
2. **Fix BUG-004 (High):** Enforce a minimum password length of at least 8 characters with a clear validation error message shown to the user.
3. **Fix BUG-001 (Medium):** After successful registration, automatically log the user in and redirect to the Dashboard. This improves onboarding UX.
4. **Fix BUG-003 (Low):** Add asterisk (`*`) markers to all required field labels on the registration form for standard UX clarity.
5. **Extend test coverage** to Firefox and WebKit browsers to ensure cross-browser compatibility.
6. **Add test.describe grouping** in the test files to better organize tests by feature (Registration, Login, Validation, etc.).